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MAJOR PROJECT REPORT

CAREERTRACK: FULL-STACK JOB PORTAL & APPLICANT TRACKING SYSTEM

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Program: Full Stack Web Development / MERN Stack Training Program
Live Frontend URL: https://career-track-eosin.vercel.app
Live Backend API URL: https://careertrack-isqj.onrender.com
GitHub Repository: https://github.com/iaditi97/CareerTrack

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1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Traditional recruitment workflows often create unnecessary friction for both job seekers and hiring teams. Many job portals force candidates to complete lengthy account sign-ups before applying, leading to high drop-off rates, while small and mid-sized employers struggle to track incoming resumes, filter applicants, and maintain structured hiring pipelines.

CareerTrack solves both challenges by providing a streamlined, dual-interface MERN-stack recruitment platform:

- For Job Seekers: Instant access to browse active job openings, filter by employment type or keywords, and submit applications with PDF/DOCX resumes without mandatory account registration.
- For Employers / Admins: A secure, JWT-authenticated dashboard to publish and manage job postings, review candidate resumes, update application pipeline statuses, record internal evaluation notes, and export applicant data to CSV.

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2. SYSTEM ARCHITECTURE & TECH STACK

CareerTrack follows a decoupled client-server REST API architecture deployed on cloud infrastructure for 24/7 availability:

- Frontend Client: React.js (Vite), Tailwind CSS, Framer Motion, React Router, Axios, and Lucide React Icons. Hosted on Vercel with SPA rewrite routing.
- Backend REST API: Node.js, Express.js, Multer (multipart file handling), Express Rate Limit, and CORS middleware. Hosted on Render.
- Database & ODM: MongoDB Atlas cloud cluster managed via Mongoose schemas.
- Security & Authentication: JSON Web Tokens (JWT) for stateless session authorization, Bcryptjs for password hashing, file type/size validation (PDF/DOCX up to 5MB), and duplicate application prevention.

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3. CORE FUNCTIONAL MODULES

3.1 Public Job Discovery & Filtering Module

- Fetches live job listings from MongoDB Atlas via GET /api/jobs.
- Supports instant keyword search (job title, company, location) and employment type filtering (Full-time, Part-time, Internship, Contract).

3.2 Frictionless Candidate Application Module

- Displays full job descriptions, salary ranges, locations, and requirements.
- Validates candidate inputs and enforces strict client- and server-side checks on resume uploads (allowing only .pdf and .docx formats under 5MB).
- Prevents duplicate submissions by checking candidate email + jobId pairs.

3.3 Employer Authentication & Dashboard Module

- Provides secure Employer registration and login (/api/auth/register and /api/auth/login) using hashed credentials and signed JWT tokens.
- Allows authenticated employers to create, toggle active/closed status, or delete job postings.
- Displays incoming applications per job with direct resume viewing links, status pipeline controls (Submitted, Under Review, Shortlisted, Selected, Rejected), internal recruiter notes, and one-click CSV report export.

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4. DATABASE DESIGN SUMMARY

The MongoDB database (job_portal) is structured into three normalized collections:

1. Admins: Stores employer account credentials (email, hashed password, role, timestamps).
2. Jobs: Stores job listing metadata (title, company, location, salary, type, description, requirements, status, createdAt).
3. Applications: Links candidates to jobs via ObjectId references (jobId, name, email, phone, qualifications, experience, resumeUrl, status, notes, appliedAt).

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5. SECURITY & DEPLOYMENT HIGHLIGHTS

- Environment Isolation: Sensitive database URIs and JWT secrets are stored in cloud environment variables on Render and excluded from Git via .gitignore.
- API Protection: Configured express-rate-limit to prevent abuse and strict CORS policies restricting write operations to the production Vercel origin.
- Continuous Deployment: Connected GitHub main branch to Vercel and Render for automated builds and zero-downtime updates.

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6. CONCLUSION & FUTURE ENHANCEMENTS

CareerTrack successfully delivers a production-ready, end-to-end hiring and applicant tracking solution. Future enhancements include automated email notifications when a candidate's status changes, cloud object storage (AWS S3 / Cloudinary) for permanent resume archiving, and AI-assisted resume keyword matching.

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